

CHETAN TANAJI RAKSHE

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[Google Scholar](#) | [LinkedIn](#) | [GitHub](#)

RESEARCH SUMMARY

Computational neuroscience researcher with extensive experience in experimental paradigm design, high-density EEG data acquisition, and advanced biomedical signal processing. Specializing in designing advanced AI/ML architecture to extract core diagnostic and biometric insights from complex neural datasets.

RESEARCH INTERESTS

Neurophysiological Signal/Image Processing (EEG/fMRI) · EEG-based Biometrics · Brain-Computer Interfaces (BCI) · Multimodal Foundational Models

EDUCATION

Degree	Institution	CGPA/%	Year
PhD, Biomedical Engineering	Indian Institute of Technology (BHU), Varanasi	9.25	2022-Present
BE, Mechanical Engineering	PCCOER, Savitribai Phule Pune University	9.56	2017-2021
Higher Secondary (HSC)	Fergusson College, Pune	83.23%	2017

RESEARCH EXPERIENCE

PhD Researcher Dec 2022 - Present
School of Biomedical Engineering, IIT (BHU), Varanasi

- **EEG-based Biometrics:** Working on an extracting robust neural representations from high-density EEG to identify individual differences in cognitive profiles. Designed EEG data acquisition paradigms in E-Prime. Collected and analysed data for around 30 subjects.
- **Biometric Innovation:** Engineered a feature distribution normalization framework that mitigated inter-session noise, yielding an accuracy improvement in authentication protocols.
- **Funding Contribution:** Drafted technical methodology and preliminary data analysis for grant proposals submitted to DRDO and ICMR.

Junior Research Fellow April 2022 - Dec 2022
School of Biomedical Engineering, IIT (BHU), Varanasi

- **Neuroimaging Pipeline:** Preprocessed online fMRI data (ABIDE I-II) extracted BOLD time series data.
- **Connectomics:** Applied sparse inverse covariance and fractal connectivity modeling to fMRI (BOLD signals) data, achieving high-precision diagnostic classification for ASD across multi-center datasets.

SELECTED PUBLICATIONS

**Bold denotes self-authorship*

Journal Articles

- **C. Rakshe et al.**, “Fine-Tuning EEG Channel Utilization for Emotionally Stimulated Biometric Authentication,” *IEEE Access*, 2025.

- **C. Rakshe** et al., “Autism spectrum disorder diagnosis using fractal and non-fractal-based functional connectivity analysis and machine learning methods,” *Neural Computing and Applications*, 2024. (**IF: 6.0**)
- V. Jain, **C.T. Rakshe** et al., “Age- and severity-specific DL models for ASD,” *Arabian Journal for Science and Engineering*, 2023.
- **C. Rakshe**, C.B. Thomas, M. Kalathe, V.S. Sreeraj, G. Venkatasubramanian, J. Thomas, R. Yuvaraj, D. Kumar, and J.F.A. Ronickom, “Multi-Paradigm Fusion for Robust EEG-based Authentication and Performance Enhancement in Challenging Tasks,” *Scientific Reports* (Under Review), 2026.

Conference Papers

- **C. Rakshe** et al., “Diagnostic Classification of ASD Using Fractal Functional Connectivity of fMRI and Logistic Regression,” *SHTI*, 2023.
- Tikaram, M. B. Ingle, **C. Rakshe** and J. F. Agastinose Ronickom et al., “Advancing ASD Diagnostic Classification with Features of Continuous Wavelet Transform of fMRI and Machine Learning Algorithms,” *IEEE ICCMMLA*, 2023.
- R. Ratnaik, **C. Rakshe** et al., “Diagnostic classification of ASD improves with structural connectivity of DTI and LR,” *SHTI*, 2023.

ACADEMIC SERVICE & LEADERSHIP

Organizing Committee Member

Feb 2026

Workshop: Introduction to Computational Neuroscience and Biology, IIT (BHU)

- Managing international registrations and technical coordination for a 5-day workshop sponsored by **SERB**.

Teaching Assistant

2022 - Present

IIT (BHU), Varanasi

- **Mentorship:** Supervised undergraduate capstone projects, guiding student teams through the full research cycle from hypothesis generation to final reporting.
- **Peer Review:** Invited reviewer for *IEEE Access* and *Neural Computing and Applications*.

TECHNICAL SKILLS

- **Acquisition & Design:** EEG Paradigm Design, High-density Recording, Clinical Data Management.
- **Signal Processing:** Artifact Removal (ICA/EOG), Time-Frequency Analysis, Functional Connectivity.
- **Advanced AI/ML:** Machine Learning, Deep Learning, Self-Supervised Learning (SSL), Contrastive Learning.
- **Programming:** Python (PyTorch, MNE, Braindecode, NumPy), MATLAB (EEGLAB, FieldTrip).

PROFESSIONAL TRAINING

- **CAMP Summer School:** Computational Approaches to Memory and Plasticity, IISER Pune, 2024.
- **BCI & Neurotechnology Spring School**, g.tec medical engineering, 2024.
- **Neuroimaging Workshop:** Experimental Design and Analysis, CBCS, University of Allahabad.

REFERENCES

- **Dr.Jac Fredo A.R.**, Associate Professor, School of Biomedical Engineering, IIT (BHU) Varanasi. *Email: jack.bme@iitbhu.ac.in*
- **Yuvaraj Rajamanickam**, Research Scientist, National Institute of Education, Nanyang Technological University. *Email: yuvaraj.rajamanickam@nie.edu.sg* .